Alessandro Mordasiewicz

U.S. Citizen
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Education

Purdue University West Lafayette, IN
M.S. Aeronautics and Astronautics January 2027 - December 2027 (Expected)
Major Concentration: Astrodynamics | *Minor Concentration:* Autonomy & Control GPA: 4.0/4.0

B.S. Aeronautical and Astronautical Engineering August 2022 - May 2026
Concentration: Astrodynamics | *Minor:* Computer Science GPA: 3.4/4.0

Relevant Coursework: Orbit Mechanics, Space Traffic Management, Spacecraft Attitude Dynamics, Applied Control in Astronautics, Applied Stochastic Processes, Optimization Methods for Systems and Control, Optimal Control and Estimation.

Interests: Space vehicle GNC, optimization, astrodynamics, mission design & analysis, dynamics simulation, software engineering.

Professional Experience

GNC Intern, Mission Design May 2026 - November 2026
Relativity Space *Long Beach, CA*
- Supported the GNC Mission Design team across algorithm development, simulation development, and analysis tasks for the Terran R medium/heavy-lift launch vehicle.
- Enhanced robustness of second stage guidance algorithm to enable accurate inertial orbit targeting (Rust).
- Developed launch window search, generation, and analysis tooling (Python).
- Built payload separation analysis tool and successfully conducted preliminary analyses (C++, Python).
- Worked in conjunction with the GNC Simulation team for several 6DoF simulation improvements (C++, Python).
- Generalized orbit maneuver formulation in trajectory optimization tooling (Julia).
- Extensively used Git (via GitLab) for version control.

Flight Mechanics Intern May 2025 - August 2025
NASA - Marshall Space Flight Center *Huntsville, AL*
- Supported the NASA HLS guidance, navigation, & mission analysis group (EV42) under the ESSCA contract.
- Developed tools to enhance lunar DEM fidelity for hazard injection and lander tip-over analysis (Python, MATLAB).
- Integrated enhanced lunar DEMs into a 6DoF landing dynamics simulation (MATLAB, Simulink).
- Verified and validated HLS landing dynamics simulations by reconstructing the Apollo 11 landing (MATLAB, Simulink).
- Developed landing visualization tool, reducing generation time from 2 hours to 2-3 minutes (Python).
- Extensively used Git (via GitHub) for version control.

Flight Dynamics Intern May 2024 - August 2024
Intelsat (now SES Satellites) *McLean, VA*
- Supported daily Flight Dynamics operations of over 70 active GEO satellites.
- Developed and maintained an interactive web application to analyze critical satellite parameters (Python, TypeScript).
- Developed data visualization/plotting tooling in operational flight dynamics software migrating to Linux (Python).
- Evaluated a GPS-based position estimation method against existing state estimation systems for GEO satellites (MATLAB).
- Gained experience with Git for version control, enabling efficient collaboration and code management.

Aerospace Engineering Intern May 2023 - August 2023
Axiom Aerospace LLC *Chicago, IL*
- Compiled FAA certification packages including comprehensive engineering reports, design drawings, and test & computations for Parts Manufacturer Approval (PMA) on FAA Part 23, 25, 27, and 29 category aircraft.
- Created and edited engineering drawings using SolidWorks and Adobe Acrobat for various aircraft parts.

Extracurriculars

Undergraduate Teaching Assistant August 2023 - May 2026
Purdue University Department of Physics and Astronomy *West Lafayette, IN*
- TA for PHYS 172: Modern Mechanics, the introductory physics course for first-year engineering students.
- Facilitated lab sections of up to 54 students, assisting students in understanding basic physics concepts and programming in Python.

Publications

R. M. Ali, A. C. Slagle, **A. Mordasiewicz**, J. Orphee, and P. J. McDonough, "Apollo 11 Lunar Module Touchdown Dynamics Reconstruction Verification and Validation for Human Landing Systems", in *48th Annual AAS Guidance, Navigation and Control (GN&C) Conference*, Breckenridge, CO, 2026, NASA Technical Reports Server, Doc. ID 20260000331 (https://ntrs.nasa.gov/citations/20260000331).

Technical Skills

Programming: Python, C/C++, Rust, Julia, MATLAB, Java, JavaScript, TypeScript (React), HTML/CSS, Bash, Git, R

Other: Simulink, GMAT, FreeFlyer, STK, Azure DevOps, GitHub, GitLab, Microsoft Office, Jira, SolidWorks, Siemens NX